

Role of Adenotonsillectomy and Tonsillectomy in Children with Down Syndrome Who Develop Obstructive Sleep Apnea by Obesity as a Risk Factor.

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Down syndrome (DS) or trisomy 21 is caused due to the presence of additional chromosome 21 in humans. DS can exist either as free trisomy 21 (nondisjunction), Robertsonian translocated DS, or as mosaic DS. Obstructive sleep apnea (OSA) is a complex condition with serious health implications for pediatric individuals with DS. OSA is common in DS, and when it is present, it appears to be extreme. Obesity and snoring are some of the OSA risk factors for children associated with DS and OSA. Adenotonsillectomy is one of the surgical protocols applied in children, which is useful in lowering the OSA in which obesity is commonly connected within normal and DS children. Tonsillectomy is the alternative procedure of surgery connected with postoperative respiratory complications, and adenotonsillectomy was found to be a safe surgical method in children and improves the quality of life. The main aim of this review is to bridge the gap between the role of OSA in normal children (46, XX/XY) and DS children (47, XX/XY+21) characterized by the presence of chromosomes and exactly what is the involvement with adenotonsillectomy and tonsillectomy when obesity is a risk factor. The treatment for OSA and obesity is rehabilitative and reversible; however, DS can be managed but not resolved because the disorder occurs from the existence of an extra chromosome during the failure of homologous chromosomal pairing separation during maternal meiosis I. This review concludes that there is a treatment for OSA and obesity and that DS children can be prevented from being obese or experiencing OSA but cannot be turned to normal chromosomes due to an extra trisomy 21. According to this review, children with DS and OSA/OSAS, as well as concomitant complications, can be treated.

Obesity, Bariatric Surgery and Obstructive Sleep Apnea-A Narrative Literature Review.

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The purpose of this review was to analyze the available literature on the subject of obesity and obstructive sleep apnea. We searched for available articles for the time period from 2013 to 2023. Obesity is listed as one of the most important health issues. Complications of obesity, with obstructive sleep apnea (OSA) listed among them, are common problems in clinical practice. Obesity is a well-recognized risk factor for OSA, but OSA itself may contribute to worsening obesity. Bariatric surgery is a treatment of choice for severely obese patients, especially with present complications, and remains the only causative treatment for patients with OSA. Though improvement in OSA control in patients after bariatric surgery is well-established knowledge, the complete resolution of OSA is achieved in less than half of them. The determination of subpopulations of patients in whom bariatric surgery would be especially advantageous is an important issue of OSA management. Increasing the potential of non-invasive strategies in obesity treatment requires studies that assess the efficacy and safety of combined methods.

Update on the aetiopathogenesis of obstructive sleep apnea: Role of inflammatory and immune mediated mechanisms.

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Obstructive sleep apnea (OSA) is often a lifestyle disease associated with obesity, which is rapidly evolving as a major health concern with diverse multisystemic implications. To prevent and mitigate its adverse effects and reduce its burden on society, its aetiopathogeneses must be precisely understood. Numerous studies focusing on the range of diverse anatomic, functional, and lifestyle factors have already been carried out to determine the possible contributory roles of these factors in OSA. Recently, evidence to validate the role of inflammatory pathways and immune mechanisms in the aetiopathogeneses of OSA is being developed. This allows for further research and translation of such knowledge for targeted therapeutic and preventive interventions in patients with or who are at risk of developing OSA.

Obesity accentuates circadian variability in breathing during sleep in mice but does not predispose to apnea.

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Obesity is a primary risk factor for the development of obstructive sleep apnea in humans, but the impact of obesity on central sleep apnea is less clear. Given the comorbidities associated with obesity in humans, we developed techniques for long-term recording of diaphragmatic EMG activity and polysomnography in obese mice to assess breathing patterns during sleep and to determine the effect of obesity on apnea generation. We hypothesized that genetically obese ob/ob mice would exhibit less variability in breathing across the 24-h circadian cycle, be more prone to central apneas, and be more likely to exhibit patterns of increased diaphragm muscle activity consistent with obstructive apneas compared with lean mice. Unexpectedly, we found that obese mice exhibited a greater circadian impact on respiratory rate and diaphragmatic burst amplitude than lean mice, particularly during rapid eye movement (REM) sleep. Central apneas were more common in REM sleep ($42 \pm 17 \text{ h}(-1)$) than non-REM (NREM) sleep ($14 \pm 5 \text{ h}(-1)$) in obese mice ($P < 0.05$), but rates were not different between lean and obese mice in either sleep state. Even after experimentally enhancing central apnea generation by acute withdrawal of hypoxic chemoreceptor activation during sleep, central apnea rates remained comparable between lean and obese mice. Last, we were unable to detect patterns of diaphragmatic burst activity suggestive of obstructive apnea events in obese mice. In summary, obesity does not predispose mice to increased occurrence of central or obstructive apneas during sleep, but does lead to a more pronounced circadian variability in respiration.

Obesity and obstructive sleep apnea: pathogenic mechanisms and therapeutic approaches.

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Obstructive sleep apnea is a common disorder whose prevalence is linked to an epidemic of obesity in Western society. Sleep apnea is due to recurrent episodes of upper airway obstruction during sleep that are caused by elevations in upper airway collapsibility during sleep. Collapsibility can be increased by underlying anatomic alterations and/or disturbances in upper airway neuromuscular control, both of which play key roles in the pathogenesis of obstructive sleep apnea. Obesity and particularly central adiposity are potent risk factors for sleep apnea. They can increase pharyngeal collapsibility through mechanical effects on pharyngeal soft tissues and lung volume, and through central nervous

system-acting signaling proteins (adipokines) that may affect airway neuromuscular control. Specific molecular signaling pathways encode differences in the distribution and metabolic activity of adipose tissue. These differences can produce alterations in the mechanical and neural control of upper airway collapsibility, which determine sleep apnea susceptibility. Although weight loss reduces upper airway collapsibility during sleep, it is not known whether its effects are mediated primarily by improvement in upper airway mechanical properties or neuromuscular control. A variety of behavioral, pharmacologic, and surgical approaches to weight loss may be of benefit to patients with sleep apnea, through distinct effects on the mass and activity of regional adipose stores. Examining responses to specific weight loss strategies will provide critical insight into mechanisms linking obesity and sleep apnea, and will help to elucidate the humoral and molecular predictors of weight loss responses.

A case-control study of obstructive sleep apnea-hypopnea syndrome in obese and nonobese chinese children.

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Obesity is a risk factor for obstructive sleep apnea-hypopnea syndrome (OSAHS) in adults. However, the prevalence of OSAHS in children is not clear, and the relationship between obesity and OSAHS remains controversial. Obese children were recruited from the endocrinology, respiratory, and ear, nose, and throat clinics. Weight-matched, age-matched, and sex-matched children were recruited as control subjects. Standard questionnaires were administered, and a standardized physical examination was carried out. Lateral neck roentgenography, sleep polysomnography, full blood count, and arterial blood gas analysis were also performed. Children with body mass index z-scores of > 1.96 were considered to be obese. An adenoidal/nasopharyngeal ratio of > 0.67 was considered to constitute adenotonsillar hypertrophy (ATH). OSAHS was defined as an apnea-hypopnea index (AHI) score of > 5 or obstructive apnea index (OAI) score of > 1 . Ninety-nine obese children and 99 control subjects were recruited into the study. Obese patients had significantly higher AHI and OAI scores, and lower sleep efficiency and minimum arterial oxygen saturation (MinSao(2)) than control subjects. The prevalence of OSAHS was significantly higher in obese children with or without the ATH groups than their nonobese counterparts (odds ratio, 1.9 vs 108, respectively; 95% confidence interval, 1.21 to 4.7 vs 6.2 to 191, respectively). Obesity, tonsillar hypertrophy, and adenoid hypertrophy were independent risk factors for OSAHS ($p < 0.001$, $p = 0.042$, and $p = 0.004$, respectively). There was a positive correlation between the degree of obesity and AHI ($r = 0.535$; $p < 0.001$), and an inverse correlation between obesity and MinSao(2) ($r = -0.507$; $p < 0.001$). End-tidal CO(2), Paco(2), and bicarbonate levels were within the normal range. Obesity is a risk factor for OSAHS, and the degree of obesity is positively correlated with the severity of OSAHS.

[Metabolic consequences of untreated obstructive sleep apnea syndrome].

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There is a recognized association between obstructive sleep apnea syndrome and metabolic syndrome, designated syndrome Z. The criteria for metabolic syndrome include at least three of the following factors: central obesity (waist circumference ≥ 102 cm for males and ≥ 88 cm for females); triglycerides ≥ 150 mg/dL; HDL cholesterol < 40 mg/dL for males and < 50 mg/dL for females; arterial blood pressure

$\geq 130/85$ mmHg; and fasting glucose ≥ 100 mg/dL. Central obesity is associated with OSAS and metabolic syndrome, and there is evidence that obstructive sleep apnea is an independent risk factor for obesity, glucose intolerance and insulin resistance. The implied mechanisms result from the activation of the sympathetic nervous system and of the hypothalamus-hypophysis-adrenal axis; activation of pro-inflammatory markers, such as IL-6 and TNF- α ; and the reduction in adiponectin levels, principally triggered by intermittent hypoxemia related to apnea. Despite such evidence, the results are controversial regarding the benefits of treating sleep apnea with CPAP in the presence of these metabolic alterations. In addition, the few studies that have addressed sleep apnea as a risk factor for dyslipidemia have presented conflicting results. Population-based, longitudinal controlled studies are necessary in order to elucidate the interaction between sleep apnea and metabolic consequences so that these individuals are properly treated.

Obstructive sleep apnea: Is it a biomarker of metabolic health in obesity.

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The prevalence of obstructive sleep apnea (OSA) increases with obesity, and OSA has been linked to increased cardiovascular risk via hypoxemia and sleep disruption. We hypothesized that if OSA contributes to cardio-metabolic risk, then 1) obese individuals with OSA will have more cardio-metabolic disease, and 2) patients with OSA who are non-adherent to CPAP treatment will have a greater incidence of cardio-metabolic abnormalities. We prospectively recruited obese patients ($n = 83$; BMI 49 ± 9 kg/m²). All patients had polysomnography and were stratified by 1) the absence/presence of OSA, and 2) metabolic health. Detailed CPAP reports were analyzed for compliance and OSA severity in 38 subjects. OSA by polysomnography was present in 69% of patients. While 79% of patients with OSA and 54% without OSA were categorized as MAO ($\chi^2 = 5.47$, $p < 0.02$), when adjusted for age, gender and BMI this difference was not significant ($p = 0.36$). Insulin levels were higher in the OSA group, but when adjusted there was no significant difference ($p = 0.350$). In patients on CPAP therapy, there was a negative associative trend between OSA control (apnea-hypopnea index) and beta-cell function (HOMA- β) ($r = -0.406$, $p = 0.076$), but no association between CPAP compliance and AHI with age, BMI, glucose, insulin, adiponectin, or insulin resistance. OSA is not independently associated with overall cardio-metabolic health and insulin resistance in obese patients, even when accounting for treatment compliance. The strongest predictors of the obese metabolic healthy phenotype in OSA patients are age, gender and BMI.

Cardiovascular Outcomes in Sleep-Disordered Breathing: Are We Under-estimating?

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Obstructive sleep apnea is a growing health concern, affecting nearly one billion people worldwide; increasingly recognized as an independent cardiovascular risk factor associated with incident obesity, insulin resistance, hypertension, arrhythmias, stroke, coronary artery disease, and heart failure. The prevalence of obstructive sleep apnea could be underestimated in the previous studies, leading to only modest predictions of cardiovascular outcomes. Using more physiologic data will increase sensitivity for the diagnosis of obstructive sleep apnea. Individuals at high risk of obstructive sleep apnea should be identified so that treatment efforts can be focused on them. This review will assess the evidence for the relationship between obstructive sleep apnea and cardiovascular consequences in the past, present, and future. We will also explore the role of adding physiological data obtained from sleep studies and its

ability to enhance the cardiovascular outcome's predictability. Finally, we will discuss future directions and gaps that need further research.

Association between metabolic score for visceral fat and obstructive sleep apnea: a cross-sectional study.

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Previous studies have established a connection between obesity and obstructive sleep apnea (OSA), novel surrogate markers of adipose accumulation may serve as more critical and reliable factors for consideration. Consequently, this study aims to explore and elucidate the correlation between metabolic score for visceral fat (METS-VF) and OSA. In this cross-sectional study, the data from the National Health and Nutrition Examination Survey (NHANES) during the period from 2013 to 2020 were adopted. Through multivariate logistic regression, restricted cubic spline regression (RCS), subgroup analyses and sensitivity analyses, the correlation between METS-VF and OSA was explored. Among 8,284 subjects, 4,176 of them were categorized as having OSA. It was observed that the quartile range of METS-VF increased, with a notable rise in the prevalence of OSA (32.8% vs. 49.8% vs. 56.9% vs. 62.1%, $p < 0.001$). Logistic regression analyses showed a significant positive correlation between METS-VF and the risk of having OSA, even after accounting for potential confounders (OR = 2.436, 95% CI: 2.065, 2.874). Subgroup analyses further revealed a stronger correlation between OSA and METS-VF among subjects who were female, younger, and Mexican Americans. RCS regression identified a positive linear correlation, without threshold effects. Sensitivity analyses with stop breathing (OR = 2.283, 95%CI: 1.169, 3.070) or snoring (OR = 2.716, 95%CI: 2.273, 3.246) as outcomes reaffirmed the positive correlation with METS-VF. Elevated METS-VF demonstrated a linear correlation with the increased incidence of OSA, suggesting the potential utility as a predictive index for OSA.
